An IoT-based Project Report on

“Parking slot Detection System”

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(Communication and Signal Processing)

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ABSTRACT

This project presents a simple and low-cost parking slot detection system using Arduino UNO, ultrasonic sensors, and an I^2C LCD display.
The system detects whether a parking slot is vacant or occupied based on the distance measured by ultrasonic sensors.
If a slot is empty, a corresponding LED glows, and the LCD shows the number of available slots.
This project demonstrates a prototype model of an intelligent parking management system that can help drivers easily identify free parking spaces and improve parking efficiency in real-world applications.

Aim

To design and implement an **Arduino-based Parking Slot Detection System** using **ultrasonic sensors**, **LED indicators**, and an **I^2C LCD display** to detect vehicle presence and display the number of vacant parking slots in real-time.

Apparatus

- Arduino UNO
- Ultrasonic Sensors
- 16x2 LCD Display(with internal I2c)
- Breadboard and Jumper Wires
- USB Cable for Programming

Theory

The system operates on the **ultrasonic distance measurement principle**.
Each ultrasonic sensor sends high-frequency sound waves that reflect when they hit a vehicle.
The **time delay** between transmission and reception is used to calculate distance:

$$\text{Distance} = \frac{\text{Time} \times 0.034}{2}$$

The Arduino processes these readings, controls the LEDs, and updates the LCD display. If the measured distance is **less than 10 cm**, the slot is considered **occupied**; otherwise, it is **vacant**.

Procedure

1. The system initializes the LCD and ultrasonic sensors.
2. Each sensor sends out ultrasonic pulses and receives the echo reflected from a nearby object.
3. The Arduino calculates the distance based on the echo time.
4. If the distance is greater than 10 cm, the slot is marked **vacant** and its LED turns **ON**.
5. If the distance is less than 10 cm, the slot is **occupied** and its LED remains **OFF**.
6. The LCD continuously displays the number of available slots and distance readings.
7. The readings update every second to reflect real-time slot status.

Code Snippets

Snippet 1: Library Initialization

```
#include <Wire.h>

#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
```

Enables I2C communication with the LCD using only SDA and SCL pins

Snippet 2: Sensor Pin Setup

```
const int trig1 = 2;

const int echo1 = 3;

const int trig2 = 4;

const int echo2 = 5;
```

Defines trigger and echo pins for two ultrasonic sensors.

Snippet 3:Distance Measurement

```
digitalWrite(trig1, LOW);
```

```
delayMicroseconds(2);
digitalWrite(trig1, HIGH);
delayMicroseconds(10);
digitalWrite(trig1, LOW);
duration1 = pulseIn(echo1, HIGH);
distance1 = duration1 * 0.034 / 2;
```

Sends an ultrasonic pulse and measures the echo return time to calculate distance.

Snippet 4:LED Control and Counting Vacant Slots

```
if (distance1 > 10) {
  digitalWrite(led1, HIGH); // Vacant
  vacantSlots++;
} else {
  digitalWrite(led1, LOW);  // Occupied
}
```

Controls LED status based on measured distance and increments the vacant slot count.

Snippet 5:LCD Display

```
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Vacant: ");
lcd.print(vacantSlots);
lcd.setCursor(0, 1);
lcd.print("S1:");
lcd.print(distance1);
lcd.print(" S2:");
```

lcd.print(distance2);

Displays the total vacant slots and current readings on the LCD screen.

Precautions

• Ensure all sensor connections are correct.
• Use a stable 5V power supply.
• Sensor Alignment should be done properly
• Keep sufficient distance between two sensors to avoid interference

Limitations and Future Applications

This system is limited to local measurements without remote monitoring.
Accuracy depends on sensor calibration and analog noise.

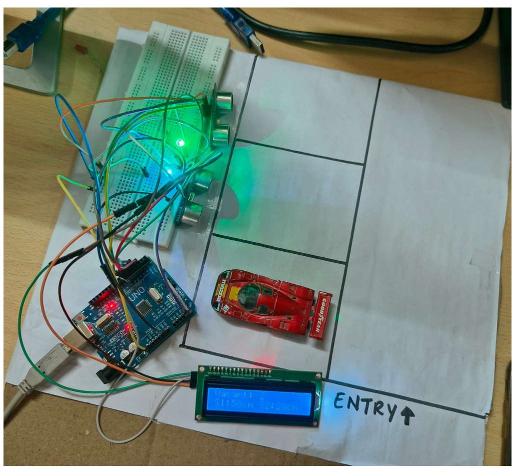
In future versions, IoT connectivity can be added to upload energy data to the cloud for analysis. Integration with Blynk or Thingspeak can allow users to view power consumption trends remotely. The project can also be upgraded to support billing estimation
and energy usage analytics.

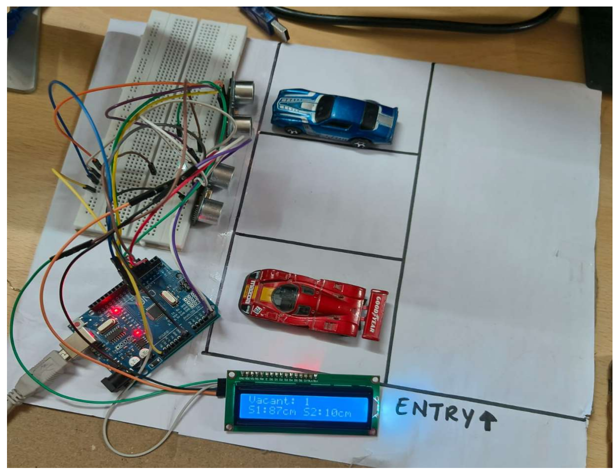
Conclusion

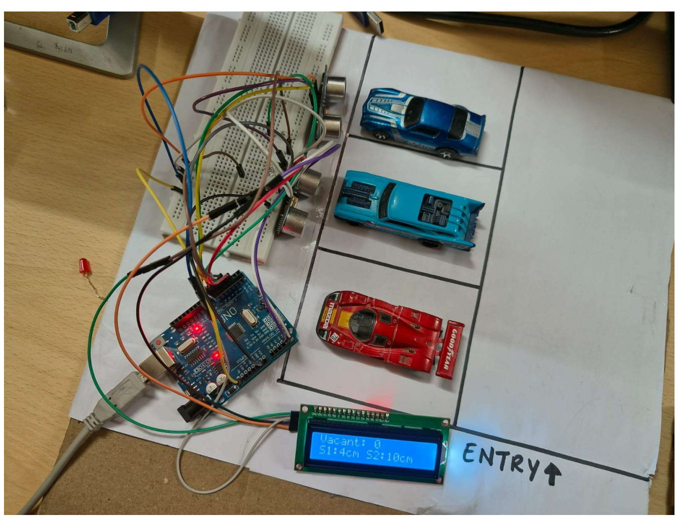
The **Arduino-based parking slot detection system** efficiently identifies the availability of parking slots using ultrasonic sensors and displays real-time status using LEDs and an I²C LCD.
Though it has certain limitations in range and environmental sensitivity, it serves as a strong foundation for developing **smart, automated parking management systems**.
With improvements such as IoT, AI, and renewable power integration, this project can evolve into a full-scale intelligent parking solution

Result

The parking detection system successfully determines whether each slot is vacant or occupied using ultrasonic sensors.
LEDs turn ON for vacant slots, while the LCD displays the number of available slots in real-time.
The system updates automatically and functions accurately in controlled indoor environments.







Appendix

Bibliography / References

• LCD Display Datasheet
• Electronics Tutorials – https://www.electronics-tutorials.ws

• Arduino Official Documentation – https://www.arduino.cc